

```
from google.colab import files
```

```
upload = files.upload()
```

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Saving aggregated_transaction.csv to aggregated_transaction.csv

Saving aggregated_user.csv to aggregated_user.csv

Saving map_transaction.csv to map_transaction.csv

Saving map_user.csv to map_user.csv

Saving top_transaction.csv to top_transaction.csv

Saving top_user.csv to top_user.csv

```
import pandas as pd
```

```
agg_trans = pd.read_csv("aggregated_transaction.csv")
agg_user = pd.read_csv("aggregated_user.csv")
map_trans = pd.read_csv("map_transaction.csv")
map_user = pd.read_csv("map_user.csv")
top_trans = pd.read_csv("top_transaction.csv")
top_user = pd.read_csv("top_user.csv")
```

```
agg_trans.head()
```

	Unnamed: 0	states	year	quarter	trans_type	trans_counts	amount
0	0	andaman-&-nicobar-islands	2018	1	Recharge & bill payments	4200	1.845307e+06
1	1	andaman-&-nicobar-islands	2018	1	Peer-to-peer payments	1871	1.213866e+07
2	2	andaman-&-nicobar-islands	2018	1	Merchant payments	298	4.525072e+05
3	3	andaman-&-nicobar-islands	2018	1	Financial Services	33	1.060142e+04
4	4	andaman-&-nicobar-islands	2018	1	Others	256	1.846899e+05

```
agg_trans.columns
agg_trans.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3594 entries, 0 to 3593
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      3594 non-null   int64
1   states          3594 non-null   object
2   year            3594 non-null   int64
3   quarter         3594 non-null   int64
4   trans_type      3594 non-null   object
5   trans_counts    3594 non-null   int64
6   amount          3594 non-null   float64
dtypes: float64(1), int64(4), object(2)
memory usage: 196.7+ KB
```

```
agg_trans = agg_trans.drop(["Unnamed: 0"], axis=1)
```

```
agg_trans.columns
```

```
Index(['states', 'year', 'quarter', 'trans_type', 'trans_counts', 'amount'], dtype='object')
```

```
agg_user = pd.read_csv("aggregated_user.csv")
```

```
agg_user.head()
```

	Unnamed: 0	states	year	quarter	brand	user_counts	percentage
0	0	andaman-&-nicobar-islands	2018	1	Xiaomi	1665	0.247033
1	1	andaman-&-nicobar-islands	2018	1	Samsung	1445	0.214392
2	2	andaman-&-nicobar-islands	2018	1	Vivo	982	0.145697
3	3	andaman-&-nicobar-islands	2018	1	Oppo	501	0.074332
4	4	andaman-&-nicobar-islands	2018	1	OnePlus	332	0.049258

```
agg_user.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6732 entries, 0 to 6731
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Unnamed: 0      6732 non-null  int64
1   states          6732 non-null  object
2   year            6732 non-null  int64
3   quarter        6732 non-null  int64
4   brand           6732 non-null  object
5   user_counts     6732 non-null  int64
6   percentage      6732 non-null  float64
dtypes: float64(1), int64(4), object(2)
memory usage: 368.3+ KB
```

```
agg_user = agg_user.drop(["Unnamed: 0"], axis=1)
```

```
agg_user.columns
```

```
Index(['states', 'year', 'quarter', 'brand', 'user_counts', 'percentage'], dtype='object')
```

```
agg_user.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6732 entries, 0 to 6731
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   states          6732 non-null  object
1   year            6732 non-null  int64
2   quarter        6732 non-null  int64
```

```

3  brand          6732 non-null  object
4  user_counts    6732 non-null  int64
5  percentage     6732 non-null  float64
dtypes: float64(1), int64(3), object(2)
memory usage: 315.7+ KB

```

```
map_trans = pd.read_csv("map_transaction.csv")
```

```
map_trans.head()
```

	Unnamed: 0	states	year	quarter	district	trans_counts	amount
0	0	andaman-&-nicobar-islands	2018	1	north and middle andaman district	442	9.316631e+05
1	1	andaman-&-nicobar-islands	2018	1	south andaman district	5688	1.256025e+07
2	2	andaman-&-nicobar-islands	2018	1	nicobars district	528	1.139849e+06
3	3	andaman-&-nicobar-islands	2018	2	north and middle andaman district	825	1.317863e+06
4	4	andaman-&-nicobar-islands	2018	2	south andaman district	9395	2.394824e+07

```
map_trans.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14636 entries, 0 to 14635
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Unnamed: 0      14636 non-null  int64
1   states          14636 non-null  object
2   year            14636 non-null  int64
3   quarter         14636 non-null  int64
4   district        14636 non-null  object
5   trans_counts    14636 non-null  int64
6   amount          14636 non-null  float64
dtypes: float64(1), int64(4), object(2)
memory usage: 800.5+ KB

```

```
map_trans = map_trans.drop(["Unnamed: 0"], axis=1)
```

```
map_trans.columns
```

```
Index(['states', 'year', 'quarter', 'district', 'trans_counts', 'amount'], dtype='object')
```

```
map_trans.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14636 entries, 0 to 14635
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   states          14636 non-null  object
1   year            14636 non-null  int64
2   quarter         14636 non-null  int64
3   district        14636 non-null  object

```

```

4  trans_counts  14636 non-null  int64
5  amount        14636 non-null  float64
dtypes: float64(1), int64(3), object(2)
memory usage: 686.2+ KB

```

```
map_user = pd.read_csv("map_user.csv")
```

```
map_user.head()
```

	Unnamed: 0	states	year	quarter	district	registered_user_counts
0	0	andaman-&-nicobar-islands	2018	1	north and middle andaman district	632
1	1	andaman-&-nicobar-islands	2018	1	south andaman district	5846
2	2	andaman-&-nicobar-islands	2018	1	nicobars district	262
3	3	andaman-&-nicobar-islands	2018	2	north and middle andaman district	911
4	4	andaman-&-nicobar-islands	2018	2	south andaman district	8143

```
map_user.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14640 entries, 0 to 14639
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Unnamed: 0            14640 non-null  int64
1   states                14640 non-null  object
2   year                  14640 non-null  int64
3   quarter               14640 non-null  int64
4   district              14640 non-null  object
5   registered_user_counts 14640 non-null  int64
dtypes: int64(4), object(2)
memory usage: 686.4+ KB

```

```
map_user = map_user.drop(['Unnamed: 0'], axis =1)
```

```
map_user.columns
```

```
Index(['states', 'year', 'quarter', 'district', 'registered_user_counts'], dtype='object')
```

```
map_user.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14640 entries, 0 to 14639
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   states                14640 non-null  object
1   year                  14640 non-null  int64
2   quarter               14640 non-null  int64
3   district              14640 non-null  object
4   registered_user_counts 14640 non-null  int64

```

```
dtypes: int64(3), object(2)
memory usage: 572.0+ KB
```

```
top_trans = pd.read_csv("top_transaction.csv")
```

```
top_trans.head()
```

	Unnamed: 0	states	year	quarter	district	trans_counts	amount
0	0	andaman-&-nicobar-islands	2018	1	south andaman	5688	1.256025e+07
1	1	andaman-&-nicobar-islands	2018	1	nicobars	528	1.139849e+06
2	2	andaman-&-nicobar-islands	2018	1	north and middle andaman	442	9.316631e+05
3	3	andaman-&-nicobar-islands	2018	2	south andaman	9395	2.394824e+07
4	4	andaman-&-nicobar-islands	2018	2	nicobars	1120	3.072437e+06

```
top_trans.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5920 entries, 0 to 5919
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Unnamed: 0      5920 non-null   int64
1   states          5920 non-null   object
2   year            5920 non-null   int64
3   quarter         5920 non-null   int64
4   district        5920 non-null   object
5   trans_counts    5920 non-null   int64
6   amount          5920 non-null   float64
dtypes: float64(1), int64(4), object(2)
memory usage: 323.9+ KB
```

```
top_trans = top_trans.drop(['Unnamed: 0'], axis=1)
```

```
top_trans.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5920 entries, 0 to 5919
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   states          5920 non-null   object
1   year            5920 non-null   int64
2   quarter         5920 non-null   int64
3   district        5920 non-null   object
4   trans_counts    5920 non-null   int64
5   amount          5920 non-null   float64
dtypes: float64(1), int64(3), object(2)
memory usage: 277.6+ KB
```

```
top_user = pd.read_csv("top_user.csv")
```

```
top_user.head()
```

	Unnamed: 0	states	year	quarter	district	registered_user_counts
0	0	andaman-&-nicobar-islands	2018	1	south andaman	5846
1	1	andaman-&-nicobar-islands	2018	1	north and middle andaman	632
2	2	andaman-&-nicobar-islands	2018	1	nicobars	262
3	3	andaman-&-nicobar-islands	2018	2	south andaman	8143
4	4	andaman-&-nicobar-islands	2018	2	north and middle andaman	911

```
top_user.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5920 entries, 0 to 5919
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Unnamed: 0            5920 non-null   int64
1   states                5920 non-null   object
2   year                  5920 non-null   int64
3   quarter               5920 non-null   int64
4   district              5920 non-null   object
5   registered_user_counts 5920 non-null   int64
dtypes: int64(4), object(2)
memory usage: 277.6+ KB
```

```
top_user = top_user.drop(['Unnamed: 0'], axis =1)
```

```
top_user.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5920 entries, 0 to 5919
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   states                5920 non-null   object
1   year                  5920 non-null   int64
2   quarter               5920 non-null   int64
3   district              5920 non-null   object
4   registered_user_counts 5920 non-null   int64
dtypes: int64(3), object(2)
memory usage: 231.4+ KB
```

Checking duplicates and null for each column

```
agg_trans.duplicated().sum()
```

```
np.int64(0)
```

```
agg_trans.isnull().sum()
```

	0
states	0
year	0
quarter	0
trans_type	0
trans_counts	0
amount	0

dtype: int64

```
agg_user.duplicated().sum()
```

```
np.int64(0)
```

```
agg_user.isnull().sum()
```

	0
states	0
year	0
quarter	0
brand	0
user_counts	0
percentage	0

dtype: int64

```
map_trans.duplicated().sum()
```

```
np.int64(0)
```

```
map_trans.isnull().sum()
```

```
      0
states  0
year    0
quarter 0
district 0
trans_counts 0
amount  0
```

dtype: int64

```
map_user.duplicated().sum()
```

```
np.int64(0)
```

```
top_trans.duplicated().sum()
```

```
np.int64(0)
```

```
top_user.isnull().sum()
```

```
      0
states  0
year    0
quarter 0
district 0
registered_user_counts 0
```

dtype: int64

```
agg_trans.to_csv("agg_trans_clean.csv", index=False)
agg_user.to_csv("agg_user_clean.csv", index=False)
map_trans.to_csv("map_trans_clean.csv", index=False)
map_user.to_csv("map_user_clean.csv", index=False)
top_trans.to_csv("top_trans_clean.csv", index=False)
top_user.to_csv("top_user_clean.csv", index=False)
```

```
from google.colab import files
```

```
files.download("agg_trans_clean.csv")
files.download("agg_user_clean.csv")
files.download("map_trans_clean.csv")
files.download("map_user_clean.csv")
files.download("top_trans_clean.csv")
files.download("top_user_clean.csv")
```


